

Faculty of Engineering & Technology
COMP433 — Software Engineering · Second Semester 2025/2026 · Section 2

Software Requirements Specification & Design

(SRSD) — Final Project Report

Cooker Shop — Online Retail System

Group 7 — Developer Group

Instructor: Dr. Adel Taweel

Team Members

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Chapter 1 — Project Planning and Management

1.1 Report Editors / Writers

Editor and assembler: Asmaa Jaraba. Each member wrote and contributed the content of their own assigned sections, which were compiled and integrated into the final report by the editor.

1.2 Business Title

Cooker Shop — Online Retail System.

1.3 Group

Group 7.

1.4 Members and University IDs

Name	University ID
Asmaa Jaraba	1231081
Misk Mashahreh	1231150
Masa Awad	1220290
Anwar Atawna	1222275
Abdel Rahman Sabbagh	1203015

1.5 Role of Each Member

- Asmaa Jaraba *: project manager.*
- Anwar Atawna: Requirement Engineer.
- Masa Awad: *technical architect*
- Misk Mashahreh: *programmer.*
- Abdel Rahman Sabbagh: Secretary

1.6 Project Management Strategy

Group 7 followed an iterative, feedback-driven process. Rather than completing the whole project in one pass, we worked phase by phase (Phase 1 through Phase 4), and after each submission we incorporated Dr. Taweel's feedback before moving to the next phase. This made our process incremental — each phase built on a corrected version of the previous one.

The team met weekly , both in person online via Google Meet. Work was divided by task ownership: each member led specific diagrams or sections, while Asmaa (project manager) and Misk acted as finalizers, reviewing and integrating everyone's work before submission, and Masa handled document assembly.

Decisions were made by discussion and consensus, with the project manager making the final call when the team could not reach agreement. We used a shared workspace [Lucidchart for diagrams, Word for documents] so members could review each other's work.

The process model most closely matches the incremental model: the system was specified, modelled, and refined in successive increments, with feedback driving each new version.

1.7 Project Manager Report

As project manager for Group 7, I coordinated our work across all four phases of the Cooker Shop project, finalized deliverables, and ensured each member's work aligned before submission.

Contributions of each member:

- **Asmaa Jaraba (Project Manager)** — Led the use-case diagram and wrote the Place Order use-case specification in Phase 3; drew the class diagram and the Place Order sequence diagram in Phase 4. Throughout the project, coordinated the team, finalized submissions, and maintained consistency across all models.
- **Misk Mashahreh** — Led the activity diagram and wrote the Make Payment use-case specification in Phase 3; produced the component diagram in Phase 4. Co-finalized deliverables with the project manager.
- **Masa Awad** — Wrote the Schedule Delivery use-case specification in Phase 3; led document assembly and the presentation deck in Phase 4, ensuring the report and slides were consistent and complete.
- **Anwar Atawna** — Wrote the Submit Maintenance Request use-case specification in Phase 3; produced the deployment diagram and the design-goals write-up in Phase 4.
- **Abdel Rahman Sabbagh** — Led the actor analysis and wrote the Manage Products use-case specification in Phase 3; drew the architecture diagram in Phase 4.

Barriers and challenges: Our main challenge was incorporating feedback correctly between phases. Several of our earlier diagrams needed significant revision — for example, our use-case diagram and class diagrams had to be reworked to follow the course's exact conventions. Coordinating these corrections across five members while keeping all diagrams consistent with one another was the hardest part of managing the project. Scheduling was also a challenge, as coordinating five members' availability around our exam period meant we had to plan meetings carefully and stay flexible with internal deadlines.

Evaluation of success: I consider the project a success. We delivered a complete and consistent set of requirements, analysis, and design models for the Cooker Shop system, and more importantly we learned to treat feedback as the standard to meet rather than a formality and how real life company works. Our models improved markedly from the early phases, and by the final phase our diagrams followed the professor's conventions closely and told one consistent story across all four chapters.

1.8 Group Members' Reports

Asmaa Jaraba:

I believe our project was successful overall. We produced a complete, well-structured set of models for the Cooker Shop system that improved significantly with each phase. As project manager, I led the use-case diagram, the class diagrams, and the Place Order sequence diagram, and I coordinated the team and finalized our submissions. My main contribution beyond my own diagrams was ensuring consistency across everyone's work — that our class diagrams, use cases, and sequence diagrams all matched and followed the course conventions.

Misk Mashahreh (Programmer):

I believe our project was successful overall. Throughout the project as a Programmer, I contributed to all four phases. In Phase 1, I developed the Users/Stakeholders and Business Capacity frameworks and conducted a comprehensive audit of the first phase. In Phase 2, I created UR8 and SR8 and completed the Effort, Time, and Cost Estimation. In Phase 3, I developed the Use Case Specification for Make Payment and designed the corresponding Activity Diagram with Swimlanes. Finally, in Phase 4, I performed the System Sequence Modeling for Make Payment and created the Component Diagram.

Masa Awad:

I believe our project was successful overall. Throughout the project, I contributed to several phases of the Cooker Shop system development. In Phase 3, I developed the Use Case Specification for Schedule Delivery & Installation (UC14), including its normal, alternative, and error flows. In Phase 4, I participated in assembling and formatting the final report, reviewing consistency between the different models and diagrams, and preparing the project presentation slides. I also contributed to organizing the final deliverables and ensuring that the documentation and presentation accurately reflected the system requirements and design.

Anwar Atawna:

I believe our project was successful overall. As Requirement Engineer, I led the Use Case Specification for Submit Maintenance Request in Phase 3, covering its normal flow, error cases, and post-conditions. In Phase 4, I built the corresponding Sequence Diagram and the Deployment Diagram, mapping our components onto the client, application server, database, and external services (Payment Gateway, Email System). Watching our models align and improve from one phase to the next was, for me, the clearest sign the project came together well.

Abdel Rahman Sabbagh:

I believe our project was successful because the group improved the Cooker Shop system across all phases by applying feedback and keeping the models consistent. I contributed to all phases through discussion, reviewing, and refinement of requirements, use cases, diagrams, and final report sections. My main responsibilities were leading the Actor Analysis, writing the Manage Product Catalog use-case specification, preparing the detailed product-management modelling work, and drawing the Architecture Diagram. I also helped ensure that the final design supports low coupling, high cohesion, and security through clear separation of services, layered architecture, and controlled system interactions.

Chapter 2 — Requirement Elicitation, Analysis and Modelling

2.1 Business Description MODIFIED

GROUP TASK — Lead: Asmaa Jaraba. Contributors: all group member discussed and reviewed.

Cooker Shop is a physical retail store specialising in the sale of cookers — gas, electric, induction, and hybrid — from a range of brands and manufacturers. The store serves households, restaurants, and small businesses, providing cooking appliances together with delivery and installation services. The business currently operates entirely manually: all sales are handled in-store or by phone, with no website or online system.

Business capacity. The store serves approximately 350 customers per month, offers around 60 cooker models, and sells about 220 units monthly, while handling multiple deliveries and installations each day. It employs 10 staff: a store manager, three sales representatives, two delivery personnel, two technicians, and two customer-support staff.

Current problems. Operating without a digital system creates several limitations: customers must visit or phone the store to ask about products; there is no online catalogue to browse or compare cookers; orders are recorded manually, risking errors or lost data; delivery and installation scheduling is poorly organised; the store cannot easily reach customers beyond its local area; there is no system to track customer orders or history; and customer support is handled manually, causing delays.

Proposed system. To address these problems, Cooker Shop requires a website system that provides an online product catalogue with full details, search and comparison functionality, online ordering, delivery and installation scheduling, secure online payment, customer accounts for tracking orders, and an administrative dashboard for managing products and orders. The system integrates with an external payment gateway for card transactions and an external email service for order, booking, and maintenance confirmations.

Stakeholders. The system's users and stakeholders include customers (individuals, families, restaurants, businesses), the store's employees, product suppliers, and the delivery and installation teams. The goal is to extend the store's reach beyond physical sales, improve the customer experience, and organise the business's internal processes.

2.2 User and System Requirements MODIFIED

GROUP TASK — Lead: Asmaa Jaraba (UR1, UR2, UR12), Masa Awad (UR3, UR4, UR5, UR6), Anwar Atawna (UR7,UR8, UR11), Abdel Rahman Sabbagh (UR9,UR10), Misk Mashahreh (UR13, UR14) **Contributors:** *Asmaa Jaraba (finalization & reviewing)*

2.2.1 User Requirements

- **UR1 — Product Browsing:** The system shall enable customers to browse the cooker product catalogue.
- **UR2 — Product Comparison:** The system shall enable customers to compare cooker products.
- **UR3 — Product Selection:** The system shall enable customers to select products for purchase.
- **UR4 — Account Registration:** The system shall enable customers to register an account.
- **UR5 — Login:** The system shall enable customers to log in to their account.
- **UR6 — Profile Management:** The system shall enable customers to manage their personal information.
- **UR7 — Online Ordering:** The system shall enable customers to place online orders.
- **UR8 — Payment:** The system shall enable customers to make payment for their orders.
- **UR9 — Order Tracking:** The system shall enable customers to track the status of their orders.
- **UR10 — Order Cancellation:** The system shall enable customers to cancel an order.
- **UR11 — Schedule Delivery:** The system shall enable customers to schedule delivery for purchased cookers.
- **UR12 — Schedule Installation:** The system shall enable customers to schedule installation for purchased cookers.
- **UR13 — Maintenance Requests:** The system shall enable customers to submit maintenance requests for purchased cookers.
- **UR14 — Product Management:** The system shall enable the administrator to manage the cooker product catalogue.

2.2.1 System Requirements

SR1 — Product Browsing

- SR1.1: The system shall display a catalogue of all available cooker products.
- SR1.2: The system shall present, for each product, its name, brand, cooker type (gas, electric, induction, or hybrid), price, and key features.
- SR1.3: The system shall display a stock status (In Stock / Out of Stock) for each product.
- SR1.4: The system shall present a product's complete technical specifications when a customer selects it.
- SR1.5: The system shall enable customers to filter the catalogue by cooker type.

SR2 — Product Comparison

- SR2.1: The system shall enable customers to select between two and four products for comparison.
- SR2.2: The system shall present a comparison of the selected products showing name, brand, type, price, and key features.
- SR2.3: The system shall enable customers to remove a product from the comparison and replace it with another without clearing the comparison.

SR3 — Product Selection

- SR3.1: The system shall enable customers to add one or more products to a shopping cart.
- SR3.2: The system shall enable customers to remove products from the cart.
- SR3.3: The system shall enable customers to change the quantity of a product in the cart.
- SR3.4: The system shall calculate and present the total cost of the items in the cart.

SR4 — Account Registration

- SR4.1: The system shall enable customers to register an account by providing their required personal details.
- SR4.2: The system shall verify that the required registration details are provided and valid before creating the account.
- SR4.3: The system shall reject registration when an account already exists for the provided email.

SR5 — Login

- SR5.1: The system shall authenticate customers using their registered credentials at login.
- SR5.2: The system shall reject a login attempt when the credentials are invalid.
- SR5.3: The system shall enable customers to recover access to their account when they have forgotten their password.
- SR5.4: The system shall reject further login attempts after a defined number of consecutive incorrect attempts and require account recovery.

SR6 — Profile Management

- SR6.1: The system shall enable customers to view their personal information.
- SR6.2: The system shall enable customers to update their personal information and saved delivery details.
- SR6.3: The system shall validate updated personal details before saving them.

SR7 — Online Ordering

- SR7.1: The system shall enable customers to place an order for the products in their cart.

- SR7.2: The system shall validate that all required order details are provided before confirming the order.
- SR7.3: The system shall verify that all ordered products are in stock before confirming the order.
- SR7.4: The system shall generate a unique order identifier for each confirmed order.

SR8 — Payment
- SR8.1: The system shall process payment for an order through an external payment gateway.
- SR8.2: The system shall record a successful payment and update the order status to "Paid".
- SR8.3: The system shall reject an order when payment is not approved and notify the customer.

SR9 — Order Tracking
- SR9.1: The system shall enable customers to view the current status of their orders.
- SR9.2: The system shall update the order status as the order progresses through processing, payment, and delivery.

SR10 — Order Cancellation
- SR10.1: The system shall enable customers to cancel an order before it has been dispatched.
- SR10.2: The system shall update the order status to "Cancelled" once a cancellation is confirmed.

SR11 — Schedule Delivery
- SR11.1: The system shall enable customers to select a delivery date and time from the available slots.
- SR11.2: The system shall assign available delivery personnel to the booking.
- SR11.3: The system shall confirm the delivery booking to the customer once it is recorded.

SR12 — Schedule Installation
- SR12.1: The system shall enable customers to select an installation appointment from the available slots.
- SR12.2: The system shall assign an available technician to the installation booking.
- SR12.3: The system shall confirm the installation booking to the customer once it is recorded.

SR13 — Maintenance Requests
- SR13.1: The system shall enable customers to submit a maintenance request for a purchased cooker.
- SR13.2: The system shall verify that the request relates to a valid previous purchase within the warranty period.
- SR13.3: The system shall generate a unique maintenance ticket identifier for each accepted request.

SR14 — Product Management
- SR14.1: The system shall enable the administrator to add a new product to the catalogue.
- SR14.2: The system shall enable the administrator to update an existing product's details.
- SR14.3: The system shall enable the administrator to remove or deactivate a product.
- SR14.4: The system shall enable the administrator to update the stock status of a product.

2.3 Effort / Time Estimation + Agreed Cost & Time MODIFIED

GROUP TASK — Lead: Misk Mashahreh. Contributors: all group members discussed, Asmaa Jaraba reviewed and edited.

Our developer team consists of 3 developers. Effort is estimated for each user requirement using the course's simplified developer-driven method. For each requirement we estimate the average effort per developer (in person-weeks, pw) and how many developers can work on it concurrently. To optimise development time we allocate developers concurrently across requirements wherever the work allows, making full use of the team rather than estimating for a single developer working alone.

pw = person-week · w = week · effort = work for one person employed full-time · schedule time = calendar time to complete, including a 30% buffer.

UR	Requirement	Effort (avg/developer)	No. of Developers (concurrent)	Total Effort (one developer)
UR1	Product Browsing	1 pw	1	= 1 × 1 = 1 pw
UR2	Product Comparison	1 pw	1	= 1 × 1 = 1 pw
UR3	Product Selection (Cart)	1 pw	2	= 1 × 2 = 2 pw
UR4	Account Registration	1 pw	1	= 1 × 1 = 1 pw
UR5	Login	1 pw	1	= 1 × 1 = 1 pw
UR6	Profile Management	1 pw	1	= 1 × 1 = 1 pw
UR7	Online Ordering	2 pw	2	= 2 × 2 = 4 pw
UR8	Payment	2 pw	2	= 2 × 2 = 4 pw
UR9	Order Tracking	1 pw	1	= 1 × 1 = 1 pw
UR10	Order Cancellation	1 pw	1	= 1 × 1 = 1 pw
UR11	Schedule Delivery	1 pw	1	= 1 × 1 = 1 pw
UR12	Schedule Installation	1 pw	1	= 1 × 1 = 1 pw
UR13	Maintenance Requests	1 pw	1	= 1 × 1 = 1 pw
UR14	Product Management	1 pw	2	= 1 × 2 = 2 pw
Total / avg	Average developers = 18 / 14 = 1.3	**16 pw**	1.3 on avg	**22 pw**

Schedule, Cost & Profit

	Minimum (average-effort basis)	Maximum (one-developer basis)
Schedule time (effort × 1.30)	16 × 1.30 = 20.8 ≈ 21 weeks	22 × 1.30 = 28.6 ≈ 29 weeks

Cost (avg salary $650/week × max schedule)	—	**$650 × 29 w = $18,850**
Profit margin (min 10% / max 30%)	Min → $18,850 × 1.10 = $20,735	Max → $18,850 × 1.30 = $24,505

Notes (addressing the feedback)

- Team size: our developer team consists of 3 developers.
- Optimising time: developers are allocated to work concurrently on requirements wherever possible, using the full team to reduce the calendar schedule rather than estimating for a single developer.
- Schedule: the minimum schedule is 21 weeks and the maximum is 29 weeks. Because the system is specified at a fine-grained level of 14 user requirements covering a complete e-commerce platform, the total effort is distributed across all of them; the minimum represents maximum parallelism and the maximum represents minimal parallelism.
- Salary: an average developer salary of $650 per week is used, reflecting a realistic market rate rather than a token figure.

Agreed Cost and Time with the Customer

Our estimated cost ranges from $20,735 (10% margin) to $24,505 (30% margin). Following discussion with our customer group (Samir Ali, Ibrahim Al Ardah, Maen Foqaha, Yahya Sarhan), both groups agreed on a final delivery time and a total cost at the upper end of this estimate:

Agreed Delivery Time	Agreed Total Cost
Approximately 29 weeks	**$25,000**

2.4 Actors Analysis MODIFIED

GROUP TASK — Lead: Abdel Rahman Sabbagh. Contributors: Asmaa Jaraba reviewed and edited; all members discussed actor consistency with the use-case diagram.

An actor is any external entity — a person or another system — that interacts with the Cooker Shop Website System to achieve a goal. Actors are divided into principal actors, who initiate business services, and secondary actors, which are external systems that support or complete system services.

Principal Actors

Actor	Description

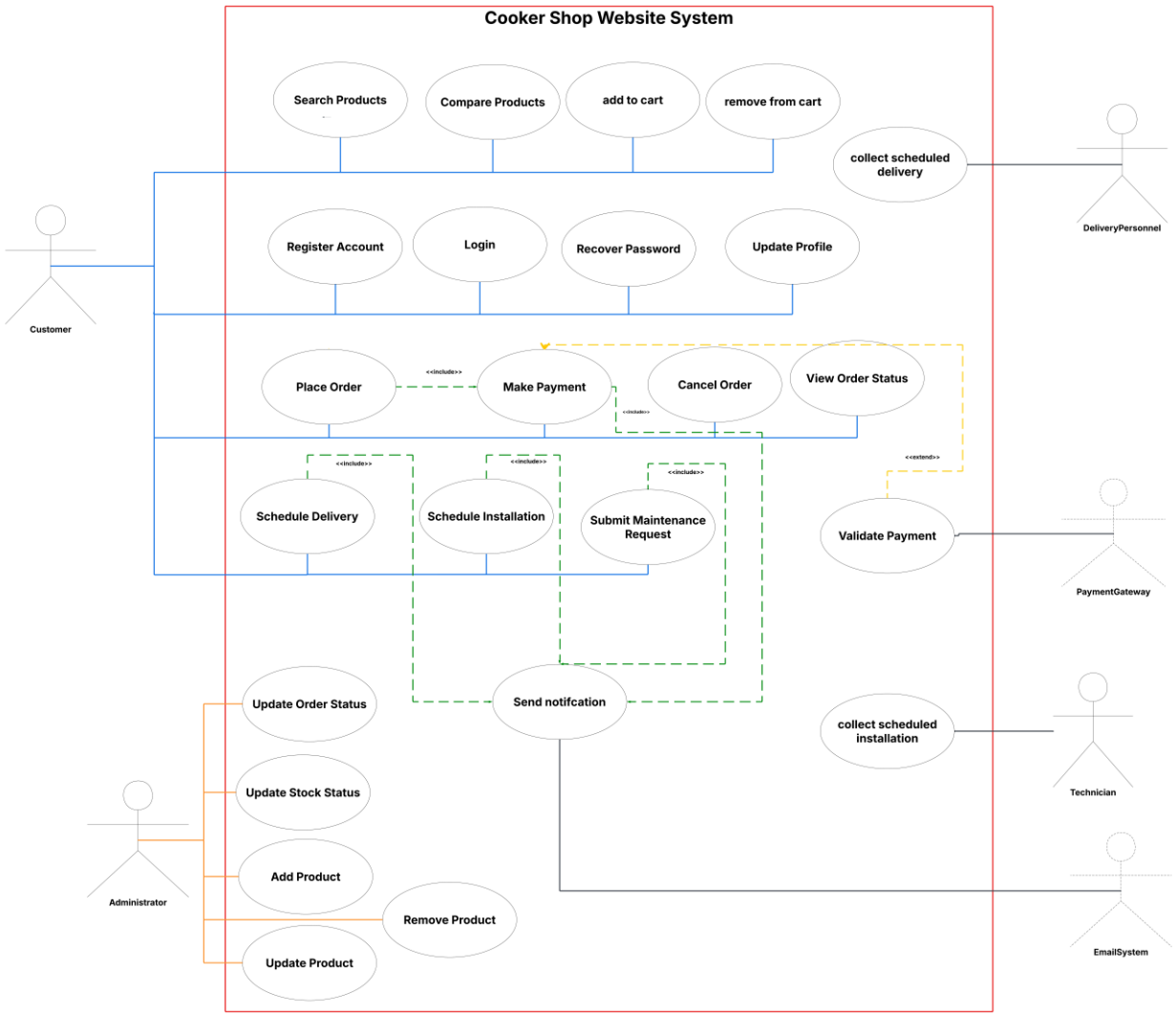
Customer	A user who browses and searches the product catalogue, compares products, manages a shopping cart, places and pays for orders, tracks and cancels orders, schedules delivery and installation, and submits maintenance requests.
Administrator	The store staff member who manages the product catalogue (add, update, and remove products), updates product stock status, and processes customer orders through the administrative side of the system.
Delivery Personnel	Store staff who collect scheduled delivery bookings from the system and carry out the delivery of purchased cookers to customers.
Technician	Store staff who collect scheduled installation appointments from the system and carry out the installation and maintenance work on cookers.

Secondary Actors (External Systems)

Actor	Description
Payment Gateway	An external service that processes and validates customers' credit and debit card transactions on behalf of the Cooker Shop, returning an approval or rejection result to the system.
Email System	An external service that sends order confirmations, booking confirmations, and maintenance acknowledgements to customers' registered email addresses.

2.5 Use-Case Diagram MODIFIED

Lead: Asmaa Jarba Contributors: Asmaa Jarba drew, rest of members reviewd



2.6 Key Use-Case Descriptions (one per member)

2.6.1. Place Order --- Asmaa Jaraba MODIFIED

System Name	Cooker Shop Website System

Use Case Title	Place Order
Actors	Customer (principal); PaymentGateway, EmailSystem (secondary)
Data	Order details, delivery address, contact number, selected products, quantities, and payment details.
Stimulus / Trigger	The customer confirms the order after reviewing the order summary.
Pre-conditions	**1.** The customer is logged in to a registered account. **2.** The customer's shopping cart contains at least one in-stock product. **3.** The final order total has been calculated and is available for confirmation.
Description	This use case describes how a registered customer confirms an order for one or more cookers. The system validates the order details, processes the payment through the Payment Gateway, generates a unique order identifier, updates stock, and sends a confirmation to the customer through the Email System.
Workflow / Flow of Events	**Normal Flow:** **1.** The customer confirms the order after reviewing the order summary. **2.** The system validates that all required order details (delivery address and contact number) are provided. **3.** The system verifies that all products in the cart are in stock. **4.** The system sends the payment request to the Payment Gateway for processing. **5.** The Payment Gateway validates the payment and returns an approval response. **6.** The system generates a unique order identifier and records the order. **7.** The system updates the stock quantity for each ordered product. **8.** The system sends an order confirmation to the customer through the Email System. **9.** The system confirms the successful order and provides the order identifier to the customer.
Alternative Flow	**4a.** Pay on delivery — at step 4, instead of online payment, the customer selects cash on delivery. The system records the order with a pending payment status and proceeds to step 6; payment is collected by the delivery personnel on delivery.
Error Flows	**2a.** A required order detail is missing or invalid — the system rejects the confirmation and returns the customer to provide the missing detail. **3a.** One or more products are out of stock — the system rejects the order, notifies the customer, and returns the affected products to the cart. **5a.** The Payment Gateway returns a rejection — the system does not record the order, sets no charge, and notifies the customer that the payment was not approved.
Post-conditions	**1.** A new order is recorded in the system with a unique order identifier. **2.** The order status is set to "Placed" (or "Paid" once payment is approved). **3.** An order confirmation has been sent to the customer. **4.** The stock quantity of each ordered product has been reduced accordingly.
Comments	The customer must be authenticated before placing an order. Out-of-stock products cannot be ordered. Full card details are not permanently stored by the system.

2.6.2. Make Payment --- Misk Mashahreh MODIFIED

System Name	Cooker Shop Website System
Use Case Title	Make Payment
Actors	Customer, Payment Gateway
Data	Card information, Order information, Customer Information, Payment Transaction information.
Stimulus / Trigger	Customer submits payment information.
Pre-conditions	1. Customer is logged in. 2. The customer has a pending order with at least one item. 3. The customer has completed the checkout process, and the order is ready for payment
Description	The customer pays for an order through an external payment gateway. The system validates the payment transaction, records the payment details, and updates the order status accordingly.
Workflow / Flow of Events	1. The customer proceeds to the payment page for a valid order. 2. The system displays the order summary, including the final order total and available payment methods. 3. The customer selects a payment method. 4. The customer enters the required payment details. 5. The customer confirms the payment request. 6. The system validates the entered payment information. 7. The system verifies that the selected payment gateway is available. 8. The system sends the payment request to the Payment Gateway. 9. The Payment Gateway processes and validates the payment. 10. The Payment Gateway returns an approval response to the system. 11. The system records the successful payment transaction. 12. The system updates the order status to **Paid**. 13. The system generates and displays a payment confirmation. 14. The system sends an order confirmation notification to the customer.
Alternative Flow	7a. If the selected Payment Gateway is unavailable, the system displays a "Payment Service Unavailable" message.
Error Flows	6a. If the entered payment information is invalid or incomplete, the system displays an error message. 10a. If the Payment Gateway rejects the payment, it returns a rejection response to the system.
Post-conditions	1. Payment transaction is recorded successfully. 2. Order status is updated to “Paid”.

	3. Order confirmation is sent to the customer.
Comments	Payment is processed through a secure external payment gateway. Full card details are not permanently stored by the system for security purposes.

2.6.3. Schedule Delivery & Installation --- Masa Awad MODIFIED

System Name	Cooker Shop Website System
Use Case Title	Schedule Delivery (UC14)
Actors	Customer, DeliveryPersonnel, EmailSystem
Data	Available delivery time slots, customer delivery address, preferred delivery date and time
Stimulus / Trigger	Customer proceeds to schedule delivery after successful payment
Pre-conditions	1. Customer has a confirmed paid order. 2. Order status is set to Paid. 3. Delivery scheduling service is available. 4. The customer's delivery address has been recorded. 5. At least one delivery slot exists in the scheduling system.
Description	This use case describes the process by which a customer schedules the delivery and installation of a purchased cooker after successful payment. The customer selects a preferred delivery date and time slot from the available options. The system validates availability, assigns suitable DeliveryPersonnel, records the booking, and sends a confirmation notification to the customer. The system ensures that delivery personnel are not double-booked for the same time slot.
Workflow / Flow of Events	**Normal Flow:** 1. The customer requests delivery scheduling for a paid order. 2. The system retrieves all available delivery and installation slots for the customer location. 3. The system presents the available slots. 4. The customer selects a preferred delivery date and time. 5. The system validates the selected slot. 6. The system checks that the slot has not already been reserved. 7. The system searches for available delivery personnel. 8. The system assigns suitable delivery personnel to the booking. 9. The assigned personnel confirm availability. 10. The system creates a delivery and installation booking record. 11. The system marks the selected slot as reserved.

	12. The system updates the order status to Delivery Scheduled. 13. The system generates a delivery confirmation notification. 14. The EmailSystem sends the delivery confirmation email to the customer. 15. The system displays the booking confirmation details. 16. **Alternative / Error Flow:** A1: Selected Slot Not Available 5a. The system detects that the selected slot is no longer available. 5b. The system retrieves alternative available slots. 5c. The system presents the alternative slots to the customer. 5d. The customer selects another slot. 5e. The use case resumes at Step 5. A2: No Delivery Personnel Available 7a. The system cannot find available delivery personnel for the selected slot. 7b. The system informs the customer that no personnel are available. 7c. The customer selects another delivery date or time. 7d. The use case resumes at Step 5.
Post-conditions	1. A delivery and installation booking record exists in the system. 2. The selected delivery slot is marked as reserved. 3. Delivery personnel are assigned to the booking. 4. The order status is updated to Delivery Scheduled. 5. A delivery confirmation notification has been generated. 6. Booking information is available for future delivery processing.
Comments	Customers may book installation simultaneously with or independently from delivery. Double-booking of the same personnel is not permitted.

2.6.4 Submit Maintenance Request --- Anwar Atawna

System Name	Cooker Shop Website System
Use Case Title	Submit Maintenance Request (UC19)
Actors	Customer, EmailSystem
Data	Invoice number, product model number, description of the issue, optional photo/video attachment
Stimulus / Trigger	The customer opens the maintenance request form and submits a request for a purchased cooker.

Pre-conditions	1. Customer is logged in to a registered account. 2. Customer has a valid invoice number from a previous purchase recorded in the system.
Description	This use case describes how a registered customer submits a maintenance request for a previously purchased cooker. The customer provides the invoice number, product model number, and a description of the issue, with an optional photo or video. The system verifies the invoice and checks that the product is still within its warranty period before accepting the request.
Workflow / Flow of Events	Normal Flow: 1. The customer enters the invoice number and product model number. 2. The customer writes a description of the issue, optionally attaching a photo or video. 3. The customer submits the request. 4. The system validates the entered information and verifies that the invoice number exists. 5. The system checks that the product is still within the warranty period. 6. The system creates the maintenance request and generates a unique ticket identifier. 7. The system sends a confirmation to the customer through the Email System. 8. The system displays a success message with the ticket identifier to the customer. **Alternative / Error Flow:** 3a. The invoice number or model number is missing or invalid, or the invoice does not exist in the system — the system rejects the submission and asks the customer to correct the information. 4b. The uploaded photo or video format is not supported — the system rejects the file and asks the customer to upload a valid file. 5a. The product is outside its warranty period — the system rejects the request and informs the customer that maintenance is covered only within the warranty period. 8a. If the product is outside the warranty period, the system rejects the request and informs the customer that maintenance requests are accepted only for products within the warranty period.
Post-conditions	Post-conditions : 1. A new maintenance request is stored in the system. 2. A unique maintenance ticket ID is generated for the request.

	3. The request status is set to "Pending Review." 4. The customer receives a confirmation email containing the maintenance ticket ID. 5. The maintenance team can view the submitted request and follow up with the customer.
Comments	Requests are accepted only for products within their warranty period.

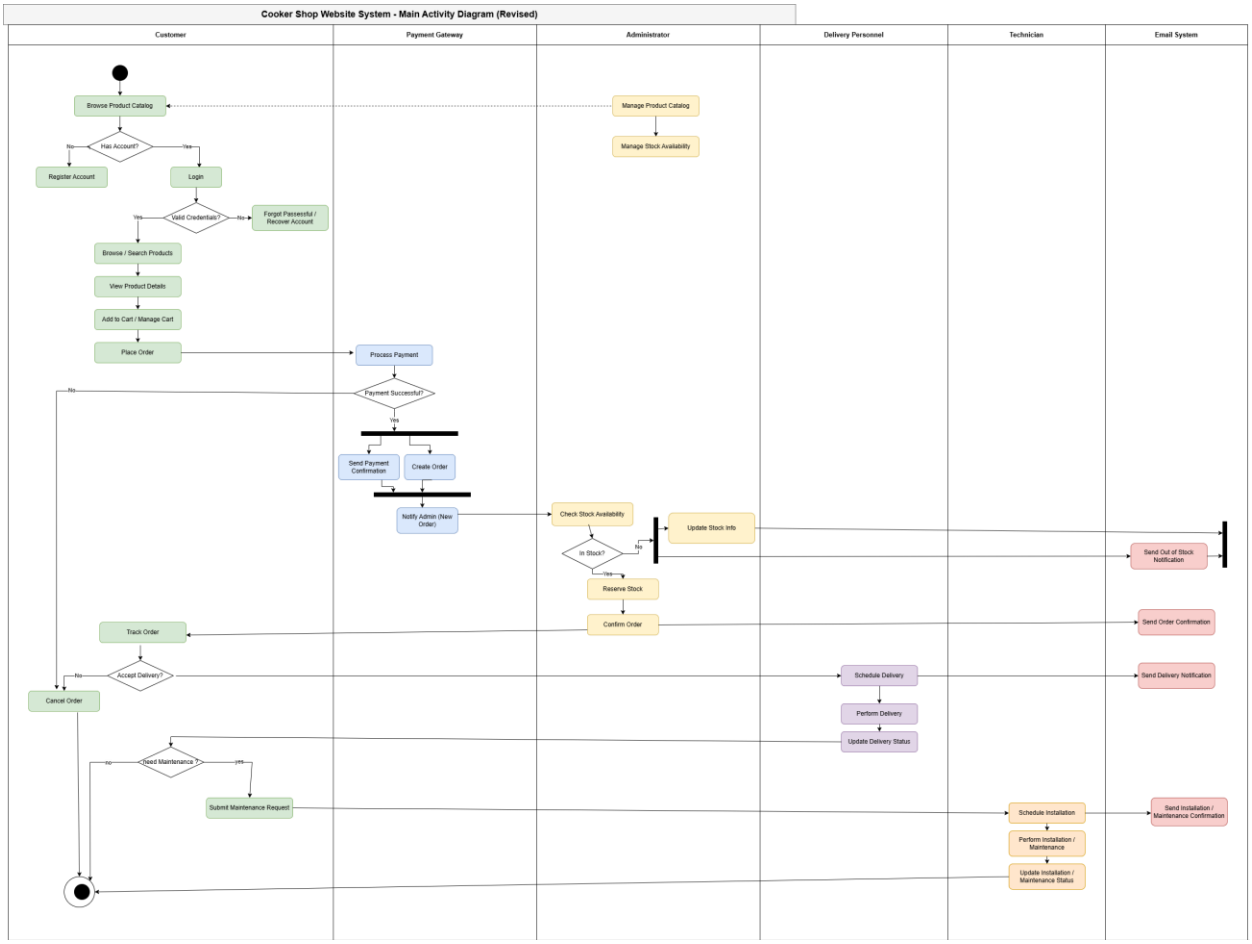
2.6.5. Manage Product Catalog--- Abdel Rahman Sabbagh

System Name	Cooker Shop Website System
Use Case Title	Manage Product Catalog (UC17)
Actors	Administrator (principal actor)
Data	Product ID/model number, product name, brand, cooker type, size, price, features, technical specifications, product image, availability status, stock status, administrator ID, and update time.
Stimulus / Trigger	The administrator needs to add a new cooker product, update an existing product, deactivate or remove a product, or change the product stock availability.
Pre-conditions	1. The administrator is authenticated. 2. The administrator has valid admin privileges. 3. The system is online and the product management service is available. 4. The product database and image storage service are available.
Description	This use case describes how the administrator manages cooker products in the Cooker Shop Website System. The administrator can add a new product, update existing product information, deactivate or remove a product, and update stock availability. The system validates the entered product data, checks duplicate model numbers, validates uploaded images, checks whether the product is linked to orders before permanent deletion, saves valid changes, updates the product catalogue for customers, and records the administrator action for audit purposes.
Workflow / Flow of Events	1. The administrator opens the Product Management page. 2. The system displays the available product management options: add product, update product, deactivate product, remove product, and update stock status. 3. The administrator selects the required operation. 4. If adding a new product, the administrator enters the product name, brand, cooker type, size, price, features, technical specifications, image, and availability status. 5. If updating an existing product, the administrator selects the product and modifies the required details.

	6. If changing stock status, the administrator selects the product and marks it as In Stock or Out of Stock. 7. The system validates that all required product data is complete and valid. 8. The system checks that the product identifier/model number is not duplicated when adding a new product. 9. The system validates and stores the uploaded product image if an image is provided. 10. If the operation is remove/delete, the system checks whether the product is linked to active or previous orders. 11. The system saves the product changes or deactivates the product according to the selected operation. 12. The system updates the product catalogue and stock availability visible to customers. 13. The system records the action with the administrator identity, action type, product ID, and update time. 14. The system displays a success message to the administrator.
Alternative / Error Flow	7a. Invalid product data: If required product data is missing or invalid, the system rejects the operation and asks the administrator to correct the missing or invalid fields. 8a. Duplicate product model: If the product identifier/model number already exists when adding a new product, the system rejects the new product and asks the administrator to enter a unique identifier/model number. 9a. Invalid image file: If the uploaded image file is invalid, unsupported, or cannot be uploaded, the system rejects the file and asks the administrator to upload a valid image. 10a. Product linked to active or previous orders: If the administrator tries to permanently delete a product linked to an active or previous order, the system prevents permanent deletion and allows deactivation instead. 13a. Audit logging failure: If the audit log cannot be recorded, the product update remains stored, but the system displays an internal warning for administrator review.
Post-conditions	1. A new product is added, an existing product is updated, or a product is deactivated/removed according to the administrator's valid action. 2. The product catalogue reflects the latest valid product information. 3. Product stock and availability status are updated. 4. Products marked as Out of Stock cannot be purchased by customers. 5. Permanent deletion does not break confirmed or historical orders. 6. The system stores a record of the administrator's action when audit logging is available.
Comments	Only authenticated administrators with proper privileges can manage the product catalogue. Product deletion should not affect confirmed or historical orders. Deactivation is preferred when a product has existing order records.

2.7 Activity Diagram MODIFIED

Lead: Misk Mashahreh Contributors: Misk Mashahreh drew.

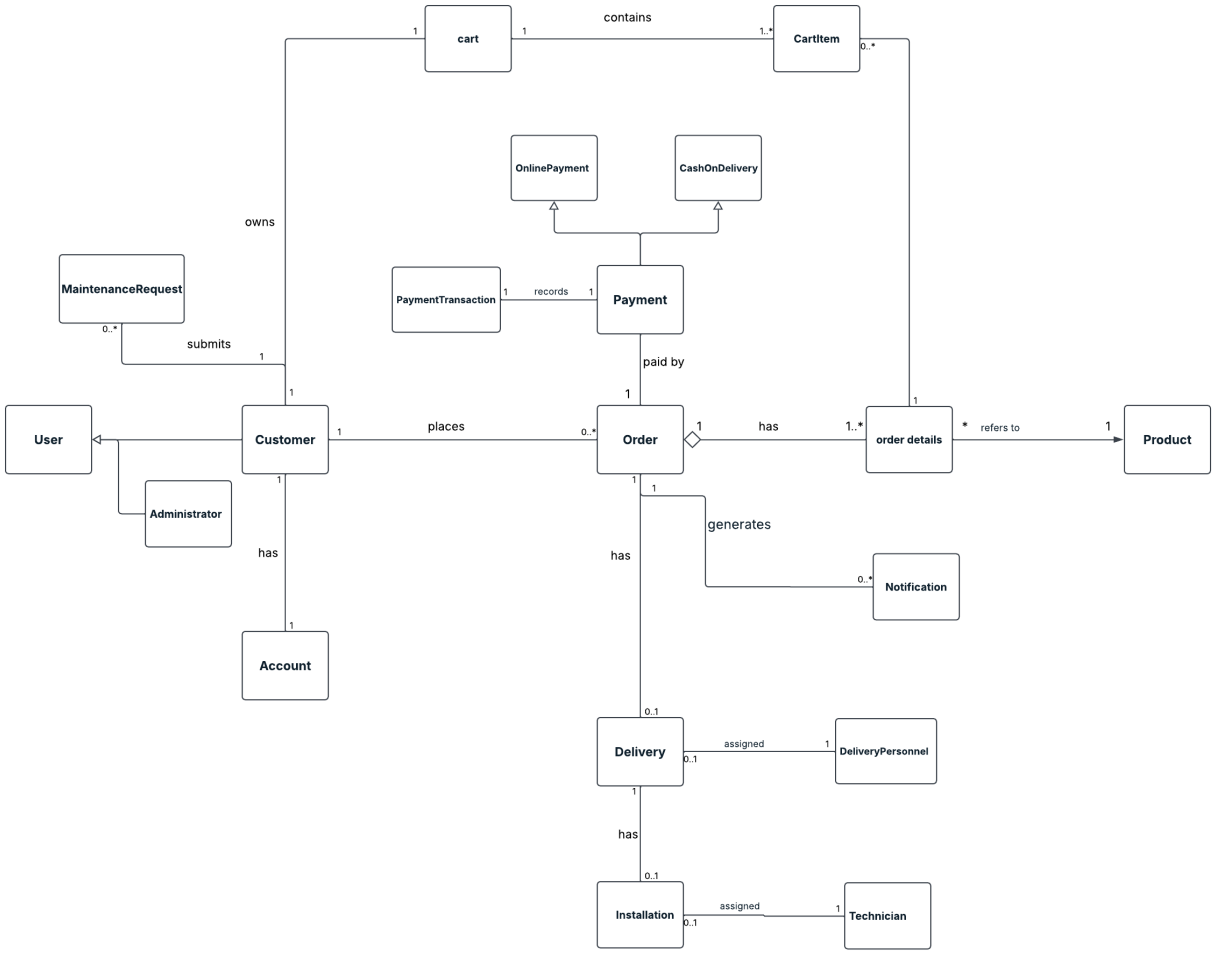


Chapter 3 — System Analysis and Modelling

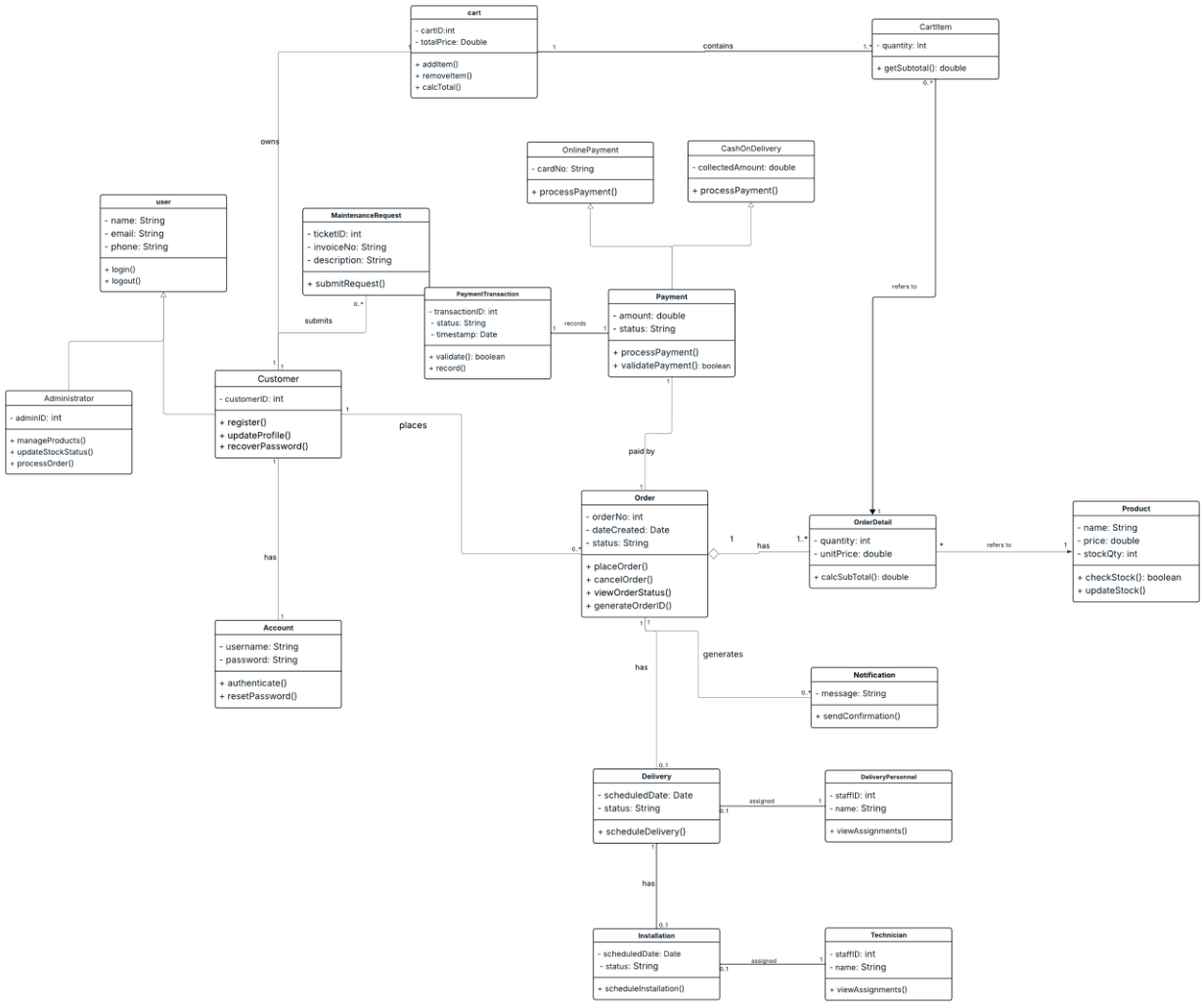
3.1 Class Diagrams (Analysis and Detailed) MODIFIED

Lead: Asmaa Jaraba Contributors: Asmaa Jarba drew, rest of members reviewd.

3.1.1. Analysis Class Diagram

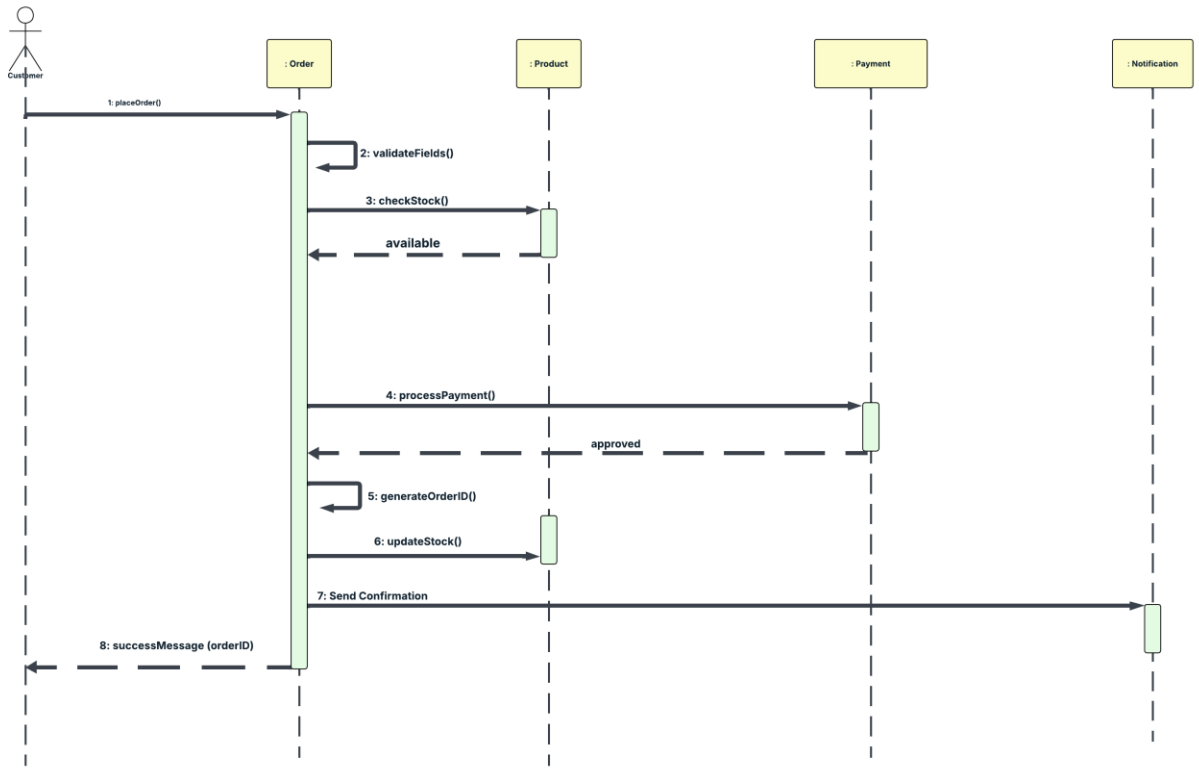


3.1.2. Detailed Class Diagram

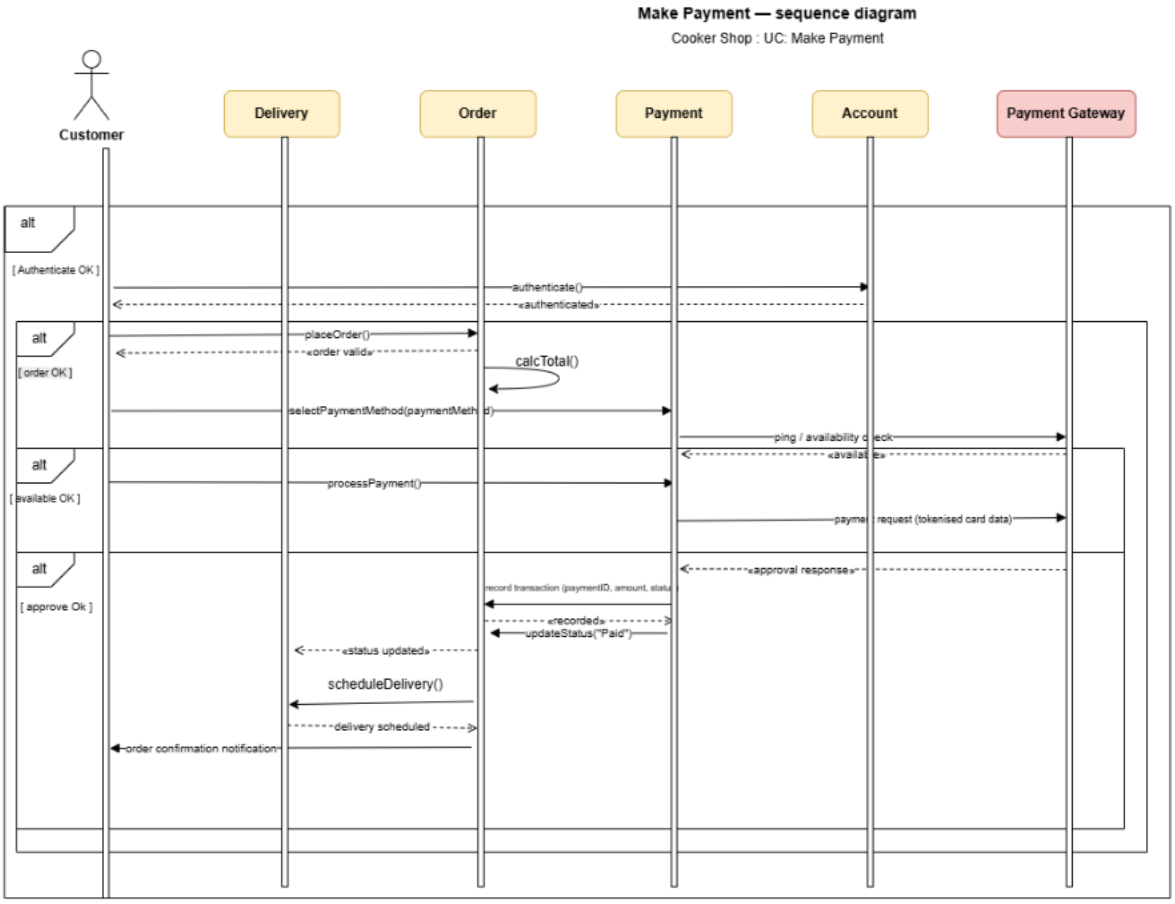


3.2 Sequence Diagrams (one per member)

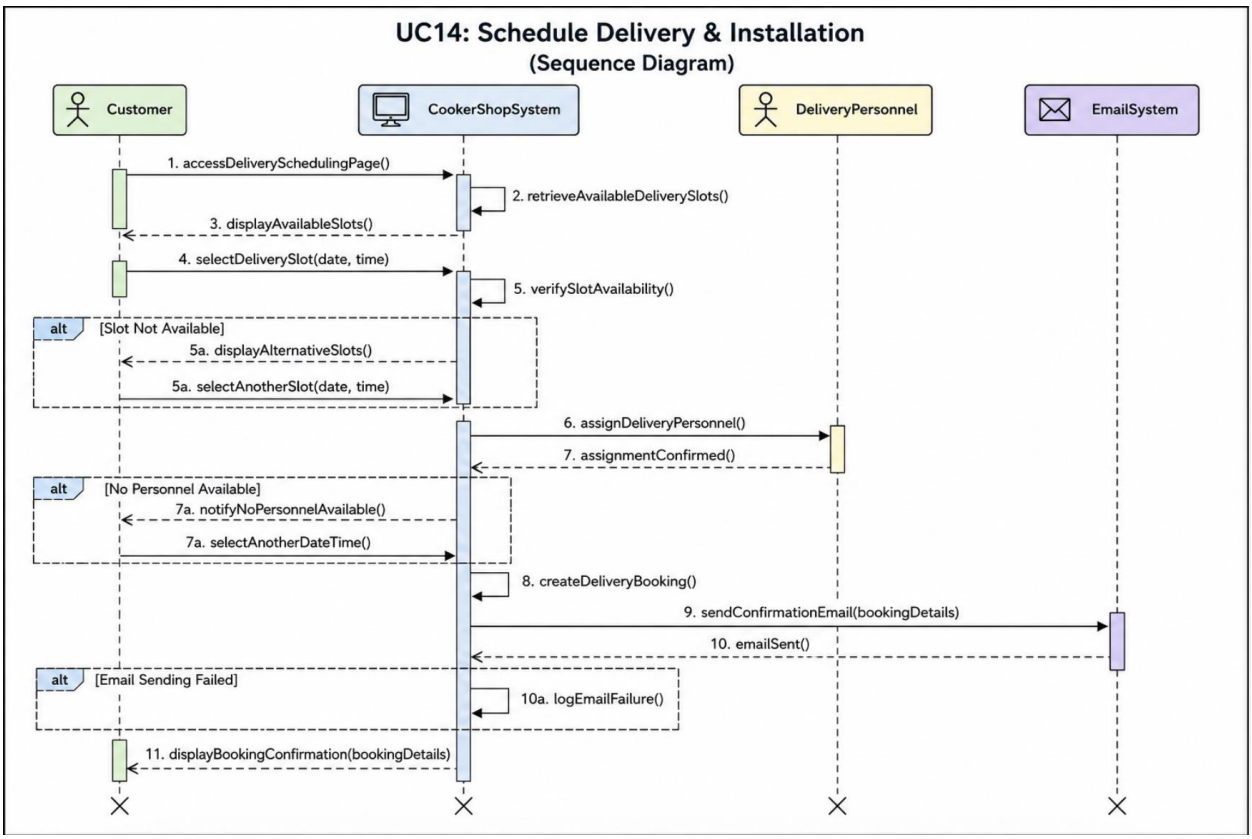
3.2.1. Place Order --- Asmaa Jaraba



3.2.2. Make Payment --- Misk Mashahreh

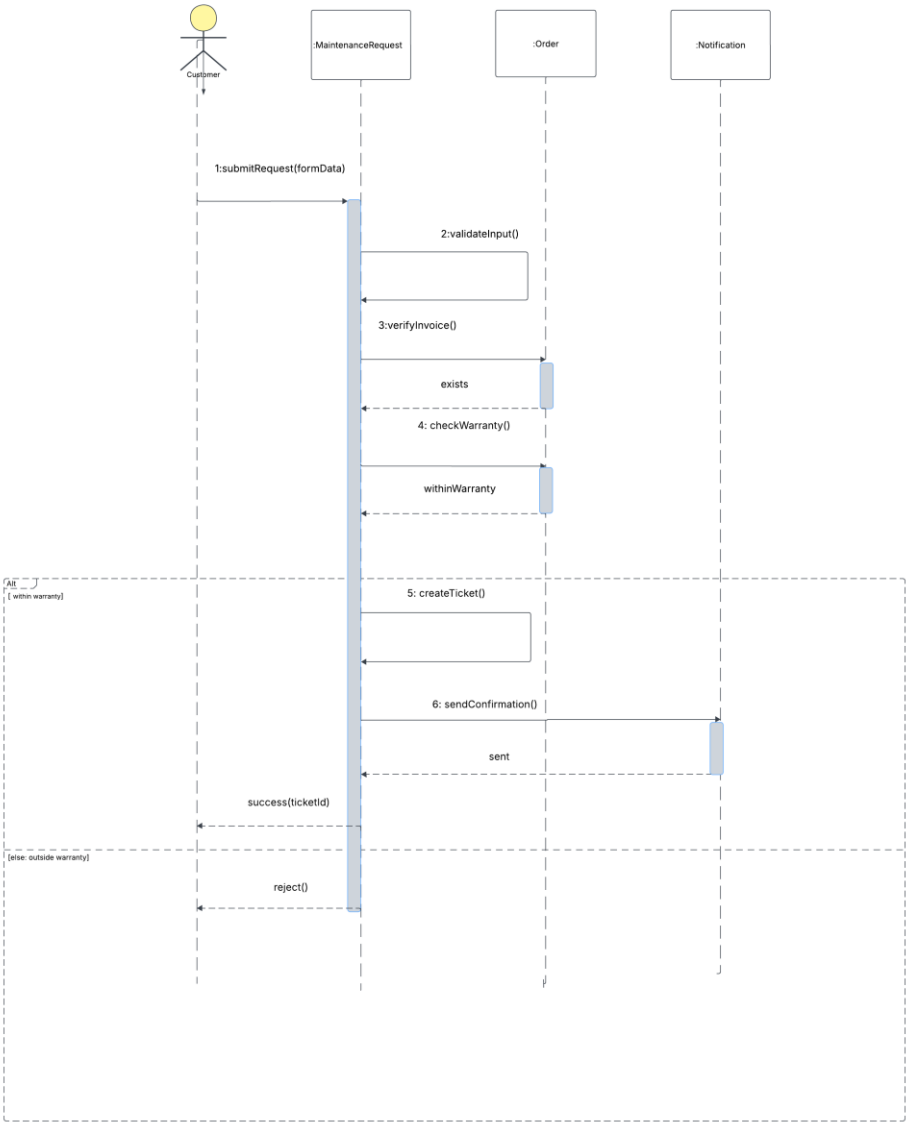


3.2.3. Schedule Delivery & Installation --- Masa Awad



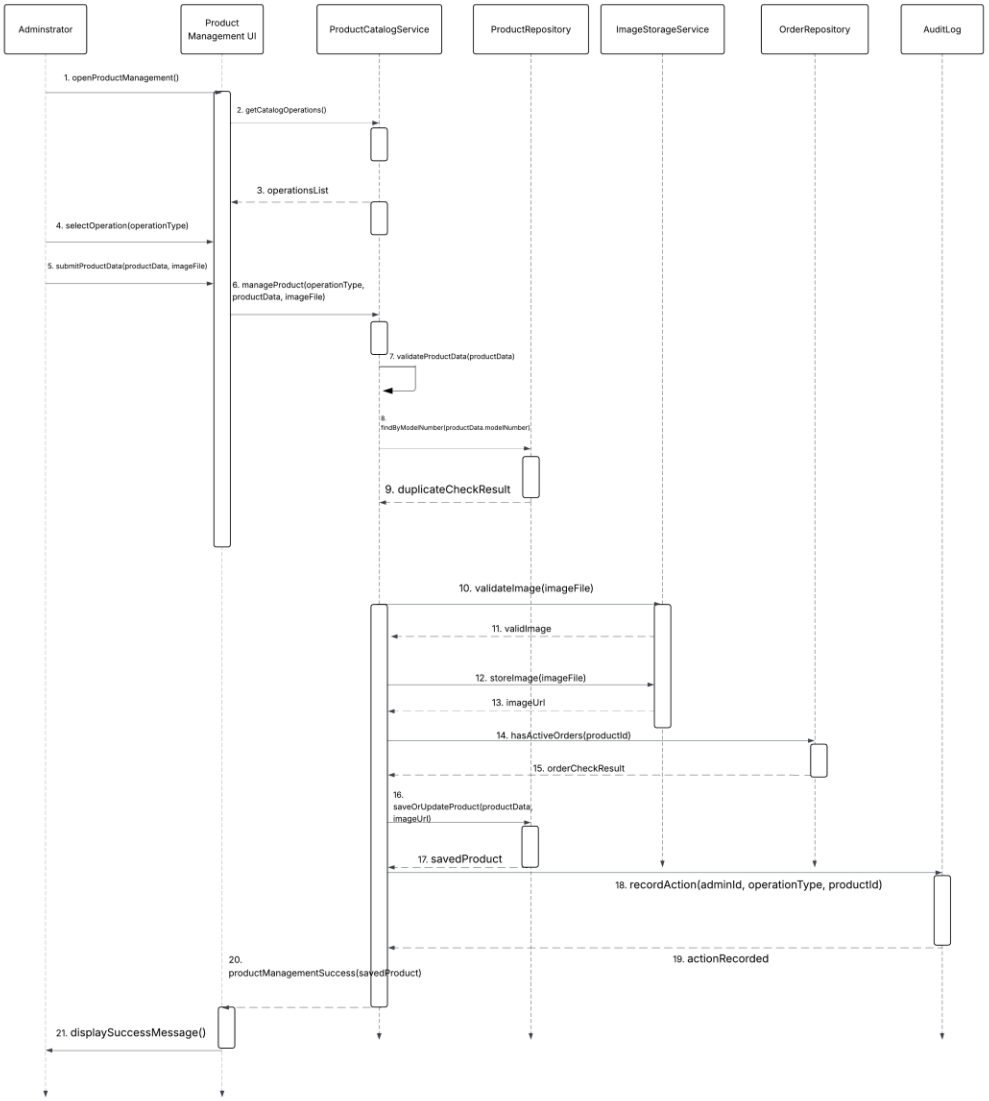
3.2.4. Submit Maintenance Request --- Anwar Atawna

Sequence Diagram — UC19 Submit Maintenance Request



3.2.5. Manage Product Catalog--- Abdel Rahman Sabbagh

Sequence Diagram — UC17 Manage Product Catalog



Chapter 4 — System Design and Modelling

4.1 Design Goals

GROUP TASK — Lead: Anwar Atawna Contributors: all members discussed and agreed on.

We selected three design goals: two general software-quality goals (Low Coupling and High Cohesion) and one non-functional goal (Security), as they directly address the main risks identified for the Cooker Shop Website System — an external payment dependency, multiple independent service domains (catalog, orders, delivery, maintenance), and sensitive customer/payment data.

1. Low Coupling

Each tier and component communicates only through well-defined interfaces, and no component accesses another tier's internals directly. For example, the Web/UI Tier never accesses the Data Tier or external services directly — all such calls pass through the Application/Business Tier via REST/API calls. This is addressed in the **Architecture Diagram** (4-tier layered client-server style) and in the **Component Diagram**, where components such as Maintenance Management and Order Processing expose narrow interfaces (e.g., `submitRequest`, `placeOrder`) rather than sharing internal data structures.

2. High Cohesion

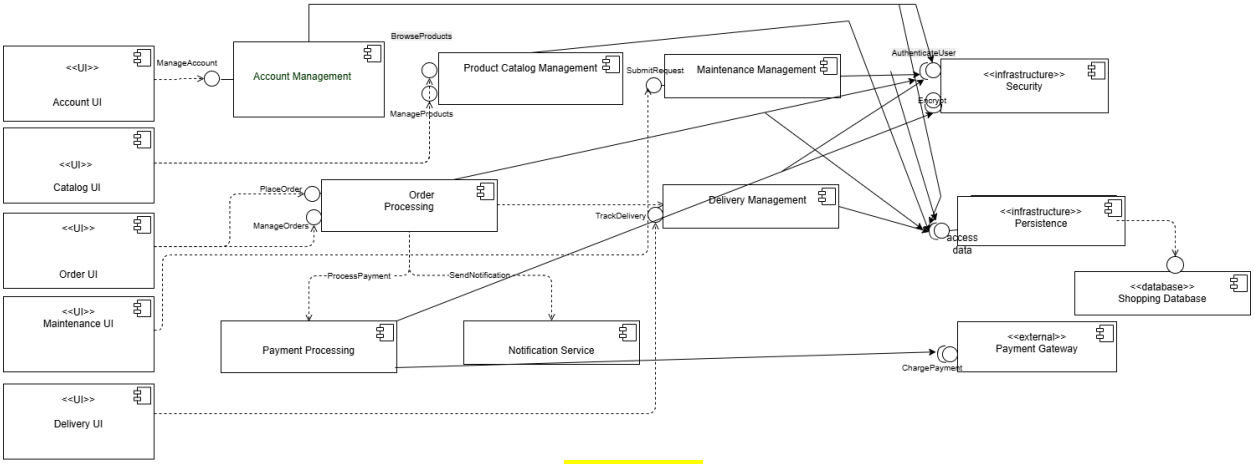
Each service is responsible for one focused business capability. Maintenance Management only handles maintenance requests and warranty verification; it does not also manage orders or payments. This separation is visible in the **Component Diagram**, where Account Management, Product Catalog Management, Order Processing, Delivery Management, Maintenance Management, Payment Processing, and Notification Service are each isolated as standalone components with a single responsibility, and in the **Detailed Class Diagram**, where `MaintenanceRequest` only contains attributes and operations relevant to a maintenance ticket.

3. Security (Non-functional Goal)

Sensitive operations must pass through authentication and authorization, and sensitive data must never be exposed unnecessarily. All external communication uses HTTPS/API, full payment card details are never permanently stored (only handled through the external Payment Gateway), and administrative actions on the product catalog are recorded in an audit log. This is addressed in the **Architecture Diagram** (Authentication & Authorization Service as a distinct component, HTTPS on all client-server links) and in the **Deployment Diagram**, where the Payment Gateway and Security Server are reached over separate, isolated channels (HTTPS, TCP/IP) rather than through the database directly.

4.2 Component Diagram MODIFIED

GROUP TASK — Lead: Misk Mashahreh. Contributors: Misk Mashahreh drew.

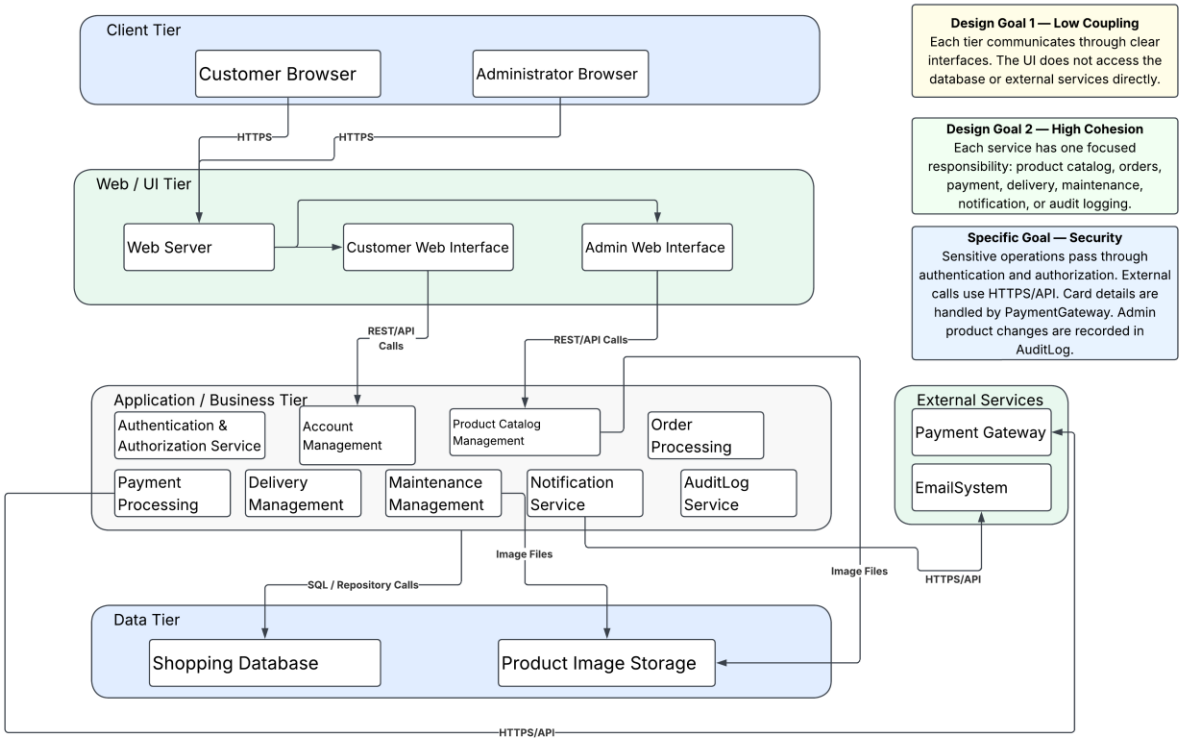


4.3 Architecture Diagram + Justification MODIFIED

GROUP TASK — Lead: Abdel Rahman Sabbagh.

Architecture Diagram — Cooker Shop Website System

Chosen architectural style: 4-tier layered client-server architecture



Architecture Justification:

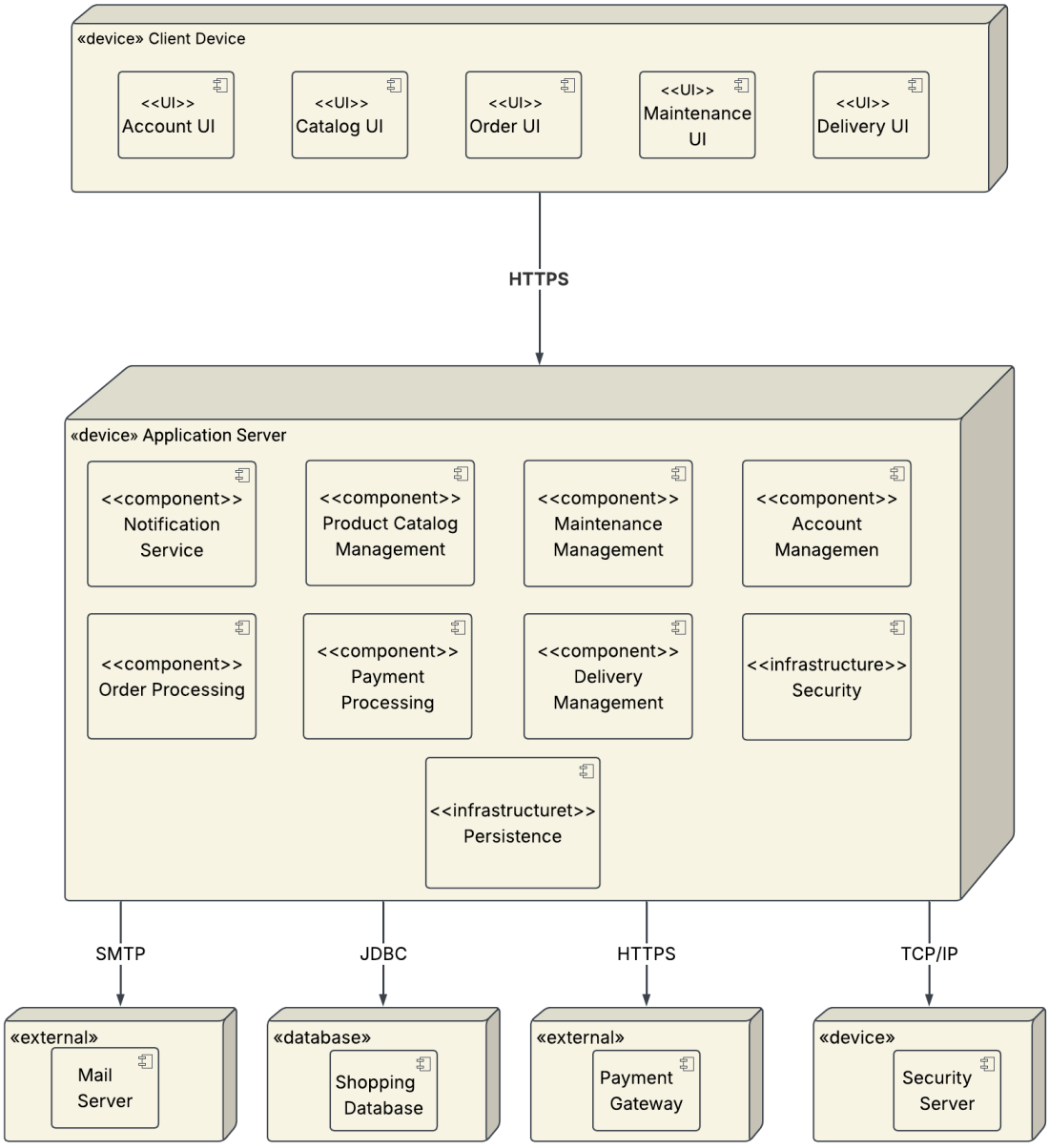
The Cooker Shop Website System uses a 4-tier layered client-server architecture. The architecture is divided into a Client Tier, Web/UI Tier, Application/Business Tier, and Data Tier, with External Services connected through controlled interfaces. The Client Tier contains the customer and administrator

browsers. The Web/UI Tier contains the web server and separates the customer interface from the administrator interface. The Application/Business Tier acts as the application server and contains the main business components: Account Management, Product Catalog Management, Order Processing, Payment Processing, Delivery Management, Maintenance Management, Notification Service, and AuditLog Service. The Data Tier stores persistent system data in the Shopping Database and stores uploaded product or maintenance images in Product Image Storage. External services, including the Payment Gateway and EmailSystem, are accessed through HTTPS/API interfaces.

This architecture was chosen because it supports the three design goals agreed by the group. It supports low coupling because each tier communicates with the next tier through clear interfaces, and the UI does not access the database or external services directly. It supports high cohesion because each application component has one focused responsibility, such as product catalog management, order processing, payment, delivery, maintenance, notification, or audit logging. It supports security because sensitive operations pass through authentication and authorization, external communication uses HTTPS/API, payment card details are handled by the external Payment Gateway rather than being stored locally, and administrator product changes are recorded through the AuditLog Service.

4.4 Deployment Diagram MODIFIED

GROUP TASK — Lead: Anwar Atawna. Contributors: Anwar Atawna drew.


«device» Client Device
<<UI>> Account UI
<<UI>> Catalog UI
<<UI>> Order UI
<<UI>> Maintenance UI
<<UI>> Delivery UI
HTTPS
«device» Application Server
<<component>> Notification Service
<<component>> Product Catalog Management
<<component>> Maintenance Management
<<component>> Account Managemen
<<component>> Order Processing
<<component>> Payment Processing
<<component>> Delivery Management
<<infrastructure>> Security
<<infrastructuret>> Persistence
SMTP
JDBC
HTTPS
TCP/IP
«external» Mail Server
«database» Shopping Database
«external» Payment Gateway
«device» Security Server

Appendix I — Minutes of Meetings

Team Meetings

Date	Attendees	Topic Discussed
3/04/2026	All Group Members	Review of Phase 1 deliverables, discussion of requirements, and planning tasks for Phase 2.
28/04/2026	All Group Members	Review of Phase 2 and Phase 3 work, discussion of use cases, activity diagrams, analysis models, and allocation of Phase 4 tasks.
8/06/2026	All Group Members	Final project review, report integration, presentation preparation, role distribution, and rehearsal for the final presentation.

Customer Meetings

Meeting 1	
Date	Saturday, 2 May 2026
Time	7:00 PM – 7:30 PM
Location	Online (Video Call)
Developer Group	Asmaa Jaraba Misk Mashahreh Masa Awad Anwar Atawna Abdel Rahman Sabbagh
Customer Group	Samir Ali Ibrahim Al Ardah Maen Foqaha Yahya Sarhan
Topics Discussed	1. General overview of the Cooker Shop business, its current manual operations, main problems, and digital goals. 2. Discussion of the core services the system must support: product browsing, ordering, payment, delivery, and installation. 3. Clarification of the four cooker types sold (gas, electric, induction, hybrid) and the approximately 60 product models available. 4. Agreement on the need for customer accounts to register, log in, track orders, and manage personal and delivery details. 5. Identification of administrator functions: product catalog management, stock control, and order processing.

	6. Discussion of after-sales support: the customer group mentioned warranty-based maintenance but did not initially provide full details. 7. UR8 (After-Sales Maintenance Requests) was identified as unclear in the original business outline provided in Phase 1. The developer group queried whether maintenance should be handled through the online system. The customer group confirmed that an online maintenance request feature is required, with warranty verification performed via the customer's invoice number. This was agreed upon and formalised as UR8.
Meeting 2	
Date	Sunday, 3 May 2026
Time	9:00 AM – 9:20 AM
Location	Birzeit UNI
Developer Group	Asmaa Jaraba Misk Mashahreh Masa Awad Anwar Atawna Abdel Rahman Sabbagh
Customer Group	Samir Ali Ibrahim Al Ardah Maen Foqaha Yahya Sarhan
Topics Discussed	1. Full review of the drafted User Requirements (UR1–UR8) with the customer group. 2. Customer group confirmed all eight user requirements accurately reflect their business needs and services. 3. Clarification that installation scheduling may be booked simultaneously with, or independently from, delivery. 4. Confirmation that maintenance requests must be tied to a valid invoice number and subject to warranty period verification. 5. Payment methods confirmed: credit/debit card online payment is required; cash on delivery is a desirable option. 6. Admin dashboard scope agreed: product management, order processing, and stock status control. 7. Final validation and sign-off on all eight requirements before proceeding to system requirements documentation.